

Statistics Lecture Notes

Meredith Shea

2026-09-22

Table of contents

Introduction	3
1 Exploring Data	4
1.1 Overview	4
1.2 Bar and Pie Charts	4
1.3 Misleading Visuals	6
1.4 Histograms	8
2 Summary Statistics	12
2.1 Measures of Center	12
2.2 Measures of Spread	14
2.2.1 Range	15
2.2.2 Inter-Quartile Range (IQR)	16
2.2.3 Outliers	17
2.3 Boxplots	18
2.4 Standard Deviation	20
2.5 Extra Practice	22
3 Examining Relationships	23
3.1 Side-by-Side Boxplots (C -> Q)	23
3.2 Contingency Tables (C -> C)	25
3.3 Scatterplots (Q -> Q)	25

Introduction

These are the lecture notes for Math 146 (34644) at Seattle Central College. These notes will be updated throughout the semester. Please note that these are built on top of your assigned readings and it is recommended that you go through those thoroughly first!

Lecture numbers will not exactly correspond to the numbers in the navigation. Here I will keep a running list on what is covered each day.

1 Exploring Data

1.1 Overview

In these notes we will look at a few common methods for visualizing data. There are two main types of data we will be concerned with in this class: **categorical** and **quantatative** data. Categorical data is when the results are descriptive (not numerical) and can be organized into a finite number of answers. Examples of categorical data include: favorite color, blood type, methods of transportation, favorite food, etc. Quantative data is data that is numerical. This can include: weights, heights, times, exam scores, etc.

The tools we use to visualize data depend on whether the data is catagorical or quantatative.

1.2 Bar and Pie Charts

Bar and **Pie charts** are two ways to visualize the distribution of **categorical** data. Suppose 250 individuals were polled about how they commute to work and the following results were obtained,

Category	Count
Car	134
Public Transit	58
Bike	44
Walk	24

We can visualize this data using either a bar or pie chart.

Creating a bar chart is simple, each **categorical variable** gets a bar where the height is represented by the number of respondents in the dataset. To create a pie chart, we start by computing the percentage of the whole for each category,

Category	Count	Percentage
Car	134	53.6%
Public Transit	58	23.2%

Category	Count	Percentage
Bike	44	17.6%
Walk	24	9.6%

The angle each pie slice gets corresponds to its percentage. For example, the *transit* category should have an interior angle that is 23.2% of 360 degrees, or

$$0.232 \cdot 360 = 83.52 \text{ degrees.}$$

Example 1.1. A class of 32 students was surveyed about whether they prefer morning, afternoon, or evening classes. The results are given in the table below.

Time of Day	Count
Morning	10
Afternoon	14
Evening	8

To compute the percentage of votes of each category, we divide the quantity by the total number of students, 32.

Time of Day	Count	Percentage
Morning	10	$10/32 = 31.25\%$
Afternoon	14	$14/32 = 43.75\%$
Evening	8	$8/32 = 25\%$

Pie Chart

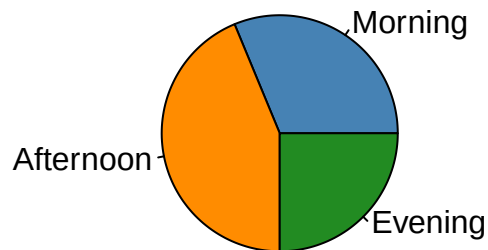


Figure 1.2: A pie chart depicting student class time preferences.

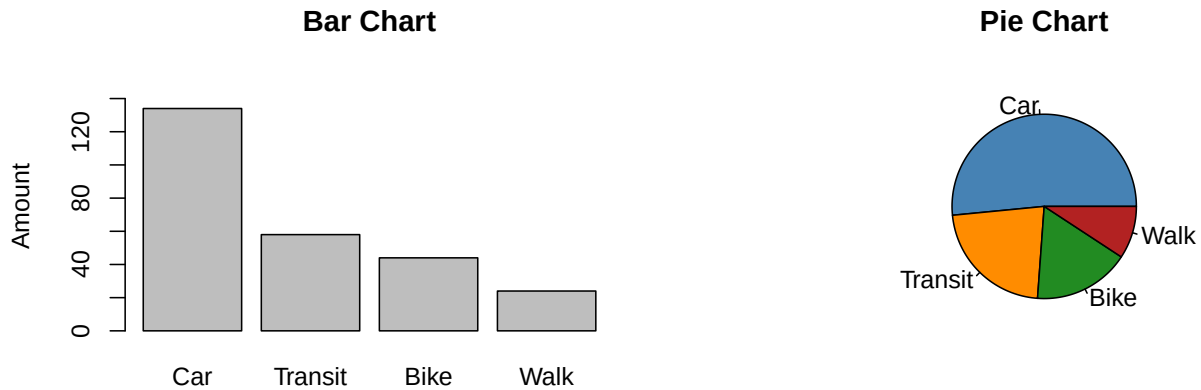


Figure 1.1: The same data shown as a bar chart (left) and a pie chart (right).

! Picking the Best Visual

In what ways are bar and pie charts similar and in what ways are they different? What are some benefits to a pie chart over a bar chart (and vice versa)?

Don't worry so much about creating a pie or bar chart by hand as we will use technology in this class. The below example explain how to create the graphs using StatCrunch.

Example 1.2 (Basic Visuals in StatCrunch). A company has collected data on 500 customers. [Open this dataset in StatCrunch](#). Use StatCrunch to create a bar chart of customer satisfaction by doing the following:

1. Under Graph, select Bar Plot, then With Data.
2. In the pop up screen, select the column SatisfactionRating. Set Type to Frequency and then hit Compute!

Next create a pie chart of the customer regions by selecting Graph > Pie Chart > With Data.

1.3 Misleading Visuals

Some visualizations can be misleading! Sometimes this is just poor judgement, but sometimes this can be done to make a specific point.

Example 1.3 (Bad Visualizations). Can you determine what makes these visuals misleading? Do you think it was done to make a specific point?

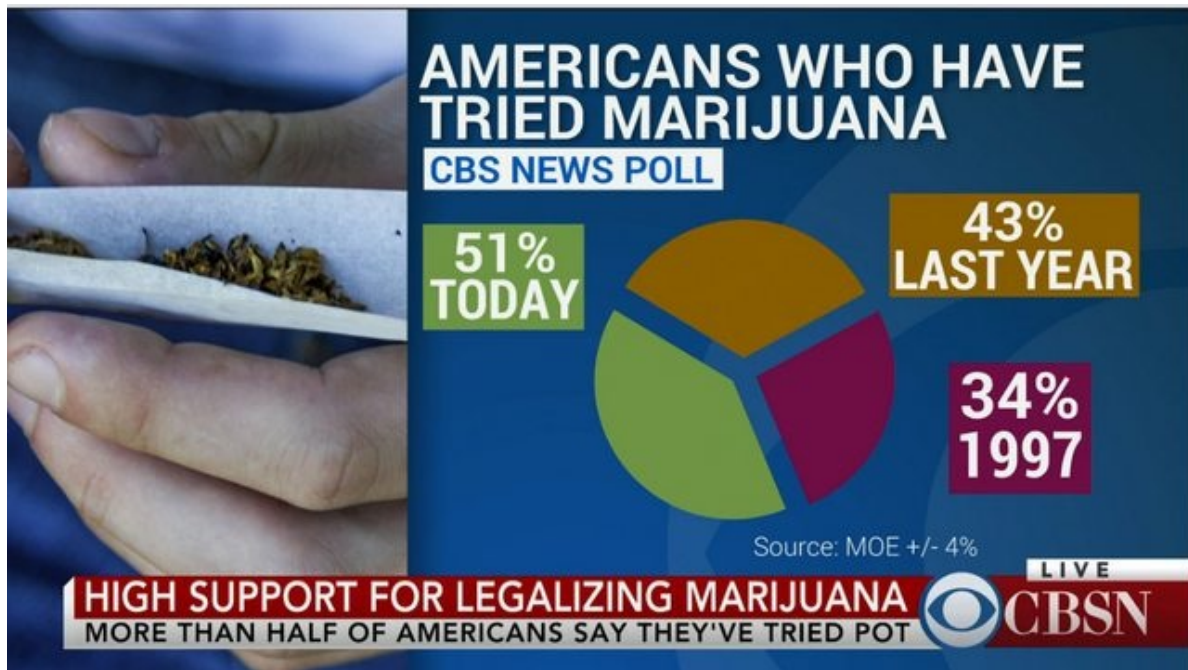


Figure 1.3: Image from www.codeconquest.com/blog/12-bad-data-visualization-examples-explained/

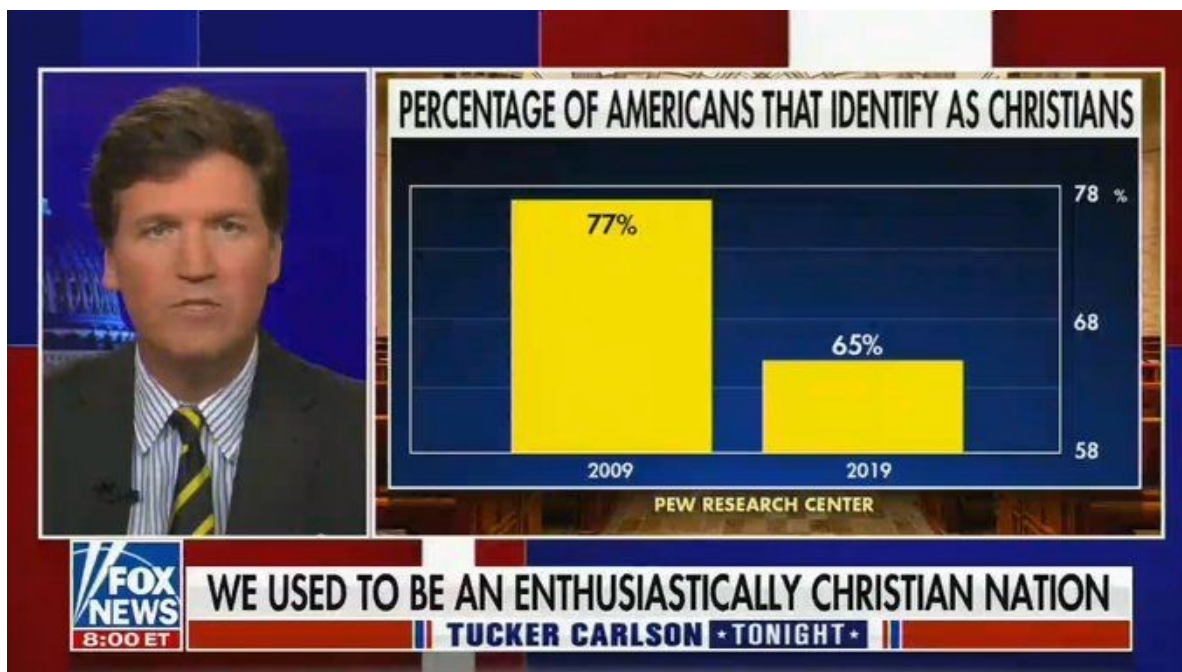


Figure 1.4: Image from www.codeconquest.com/blog/12-bad-data-visualization-examples-explained/

1.4 Histograms

Bar and pie charts are a great way to visualize *categorical* data. Now we want to address ways to visualize **quantitative** data, or data that is in the form of numbers.

Example 1.4 (Quantitative Data). Some example of quantitative data include: test scores, age, height, weight, (systolic) blood pressure, etc.

A good way to look at quantitative data is to use a **histogram**. A histogram breaks the range of all possible values into smaller intervals, then counts how many observation fall into each window. This looks kind of like a bar chart, but the bars sit right next to each other.

Example 1.5. Students in a calculus class obtain the following scores on the midterm exam,

100, 85, 91, 76, 88, 82, 95, 62, 84, 79, 90, 73, 87, 81, 92, 77, 86, 83, 80, 89, 74, 93, 85, 79, 82

The lowest score is a 62 and the highest score is 100. Let's break down the range $[60, 100]$ into intervals of length 5.

Interval	Count
(60,65]	1
(65,70]	0
(70,75]	2
(75,80]	5
(80,85]	7
(85,90]	5
(90,95]	4
(95,100]	1

From this we can make a histogram.

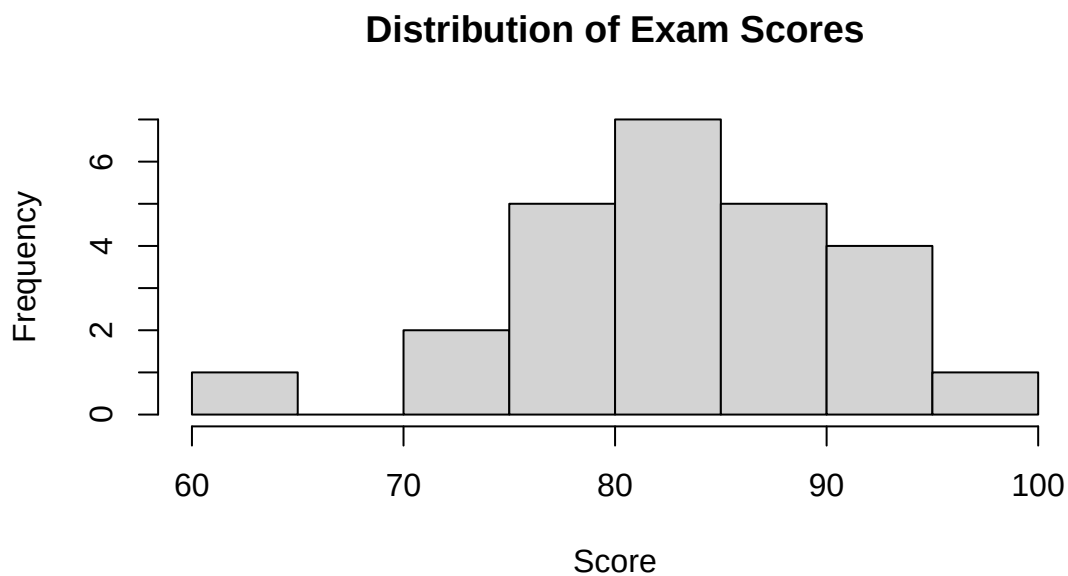


Figure 1.5: A histogram of test scores.

It is typical to call the intervals of a histogram **bins**, although the reading on WAMAP will call them **classes**. A histogram should always have bins of the same length. It can be misleading to use different bin lengths in the same way it is misleading to use breaks on the vertical axis of bar charts.

Most computer programs will set automatic bin sizes based on the distribution of the data you get it, although you will typically have an option to manually input bin specifications. In StatCrunch you are give the optional inputs of the starting value and the bin width.

! Bin Endpoints

For whatever computer program you use, make sure you know how the **endpoints** of your bin intervals are treated. In Example 1.5, I wrote intervals like $(70,75]$ which means 70 is excluded from the bin and 75 is included in the bin. The statistical programming language R (what generates the graphs in these notes) uses right inclusive bins, however StatCrunch uses *left* inclusive bins.

The histograms below represent *the same* data, just with different bin sizes.

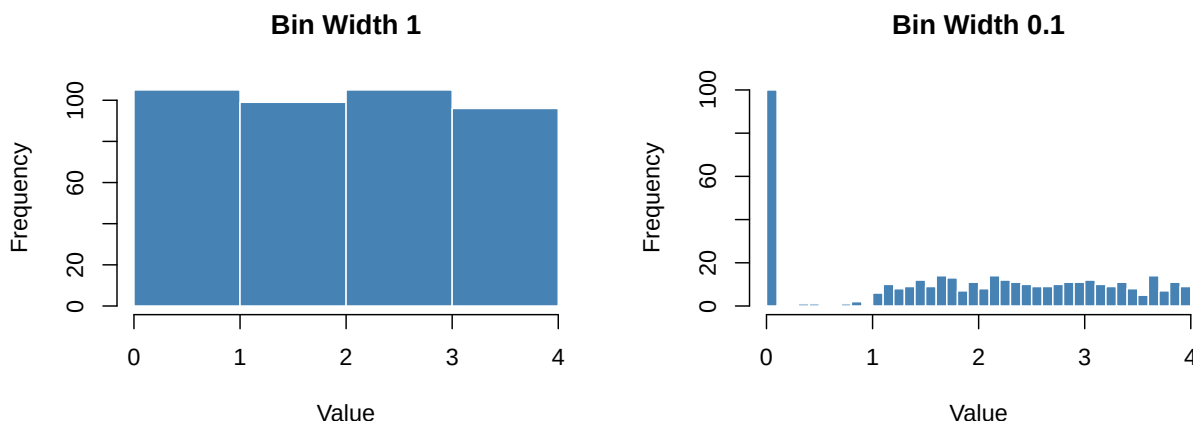


Figure 1.6: Histograms of the same dataset with different bin widths.

From the histogram on the left the data looks *uniform* across the values 0-4. When we choose a small bin width though, we see a different behavior emerge.

Example 1.6 (Histograms on StatCrunch). [Reopen the customer dataset in StatCrunch](#). Create a histogram of customer ages by going to Graph > Histogram.

Example 1.7 (High Tide Heights). The following histogram shows high tide heights over the month of July at a beach.

The number of high tides that were more than 2 meters is given by the total in the last two bins, $6 + 1 = 7$. To compute the percentage of high tides that were more than two meters, note that the total number of high tides is 62.

$$\frac{7}{62} \cdot 100 \approx 11.29\%$$

! Information Lost

What kinds of information do we lose by only seeing a histogram of the data?

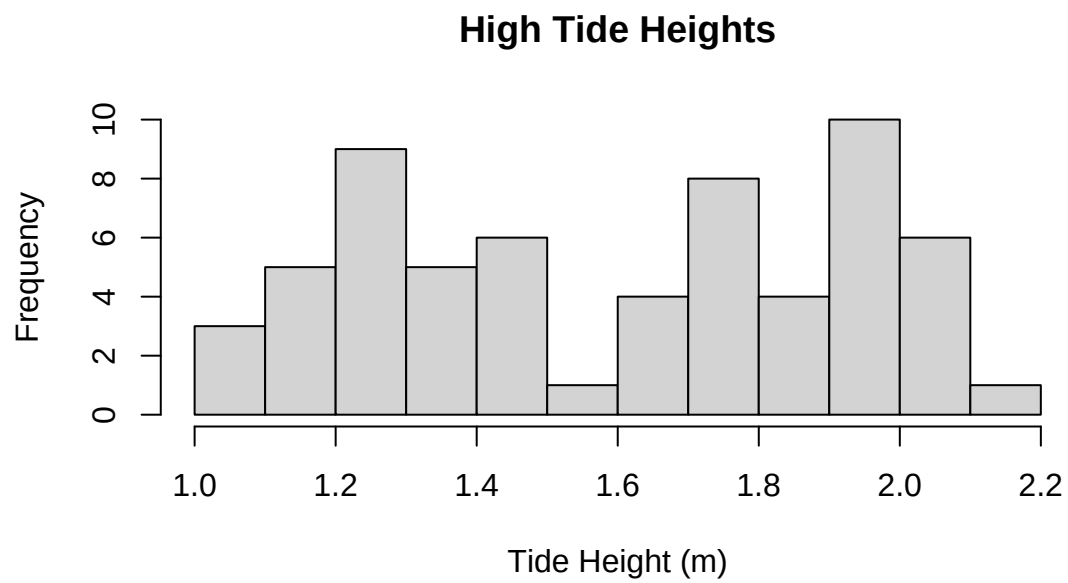


Figure 1.7: Histograms of high tide heights in July.

2 Summary Statistics

We've given a few examples of ways to visualize data, now we want to talk about some important **summary statistics**. Summary statistics are quantitative values one can use to succinctly give an idea for the shape of the distribution. Some questions we try to answer using summary statistics:

1. What is a typical value? What range does a typical value fall between?
2. What is the range of possible values? What is the range of *likely* values?

In this section we will cover different measures of **center** and measures of **spread** to help answer these questions.

2.1 Measures of Center

There are three measures of center we will cover:

- **Mode:** The value(s) with the highest frequency in the dataset.
- **Mean:** The average of the values. If a dataset has n values $x_1, x_2, \dots, x_n$, then the mean $\bar{x}$ is given by

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}.$$

- **Median:** The midpoint of the distribution. It is the number such that half the observations fall above and half fall below.

Example 2.1. A college class has 22 students with the following ages: 18, 22, 21, 21, 21, 18, 20, 42, 23, 44, 18, 18, 21, 18, 22, 18, 22, 23, 20, 38, 20, 34. To compute the mode and median it helps to reorder the data,

18, 18, 18, 18, 18, 18, 20, 20, 20, 21, 21, 21, 21, 22, 22, 22, 23, 23, 34, 38, 42, 44

The mode is 18, which occurs in the dataset 6 times.

Since the dataset has size 22 count out (in order) to the 11th and 12th observations. The median is the average of these two “middle” observations. Since they are both 21, the median of the dataset is 21.

Using the formula in the definition, the mean is $522/22 = 23.73$.

Notice that the mean is slightly larger than the median this is because there are a handful of older students with *outlier* ages. The mean gets pulled by outliers more than the median does.

[See this data on StatCrunch to perform the computations there.](#)

! Median on Datasets of Even Size

For this course, to compute the median when the dataset has an *even* number of observations average the two observation in the middle. This is a typical convention chosen in statistics.

As mentioned in Example Example 2.1, differences in the median and mean can reveal information about the shape of the distribution. A distribution is **skewed** if it has a long tail to one side.

Household Income Distribution

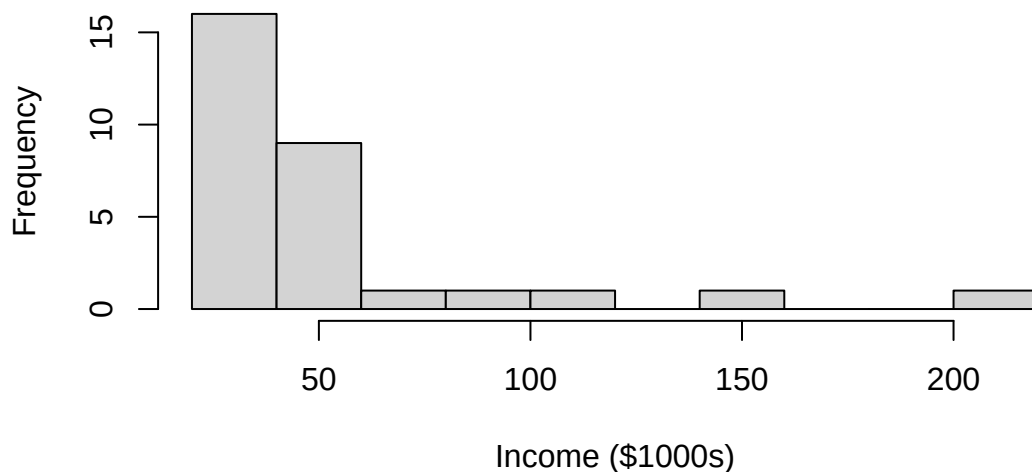


Figure 2.1: Right-skewed household income distribution.

The above dataset is skewed *to the right* since it has a long tail on the right side. For a distribution that is skewed to the right (left) the mean will be greater (less) than the median.

! Mean Versus Median

The reason **outliers pull the mean more than the median** is because mean minimizes the error *squared*, while the median minimizes the absolute error. Squaring the error means outliers will have extra impact. This fact is outside of the scope of this course, but I am happy to chat about it for curious minds!

Example 2.2 (Creating Data). Using StatCrunch complete the following:

1. Create a dataset of 10 student ages where the median is 20.
2. Is it possible to do this and have no student be 20 years old?
3. If we increased the size of the dataset to 15 students can you still create a dataset with median 20 *and* no students are 20 years old?

Example 2.3.

1. Using StatCrunch, create a dataset of 30 non-negative numbers where the mean is less than the median. Plot the histogram of the data.
2. (Challenge) If I have a dataset of 30 non-negative numbers and the median is 50, what is the smallest possible mean?

! When to use mode?

What might be a situation where the mode is the most relevant summary statistic to use?

! Nobody's Average

In 1950 the airforce measured 140 physical dimension of 4063 pilots and designed the cockpit to match the average (mean) dimensions. Lieutenant Gilbert S. Daniels then compared this “average pilot” to individual pilot records to show that no pilot was within 15% of the average on the ten most relevant dimensions. In other words, the cockpits were designed to fit no one! This led to the adaptation of adjustable seats.

2.2 Measures of Spread

Example 2.4 (Commute Times). [Data on StatCrunch](#). The below histograms depict the commute time of 500 people who work in Town A and Town B. How are the datasets similar and how are they different?

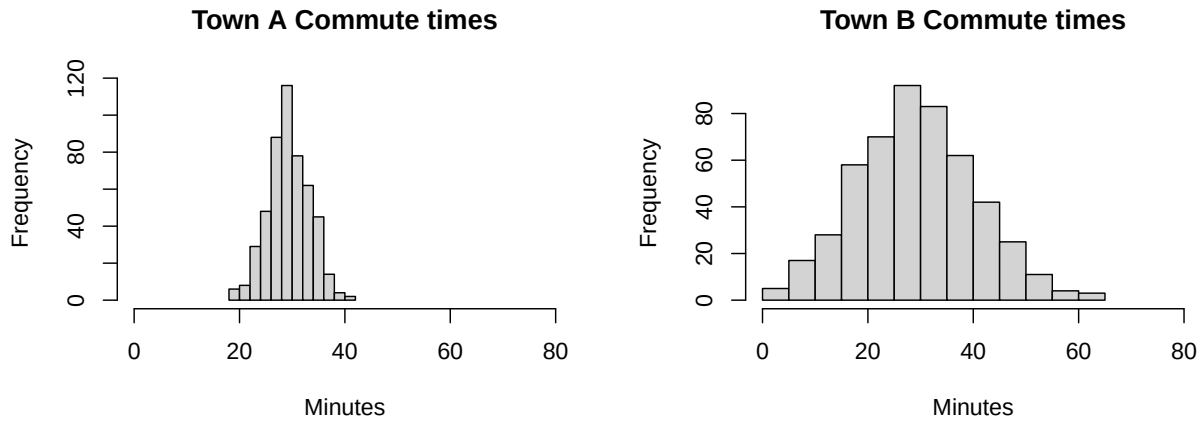


Figure 2.2: Commute times in two different towns.

The above example illustrates that it is important to have statistics that describe the *spread* of a distribution. There are a few we will investigate today.

2.2.1 Range

The **range** of a dataset is the largest observation minus the smallest observation.

Example 2.5. [The commute times data is given here.](#) Compute the range of the commuting data for each town.

Range does give us an initial look at the spread of the distribution, but outliers can obfuscate how range relates to spread. Consider two classes with the following histograms of exam scores:

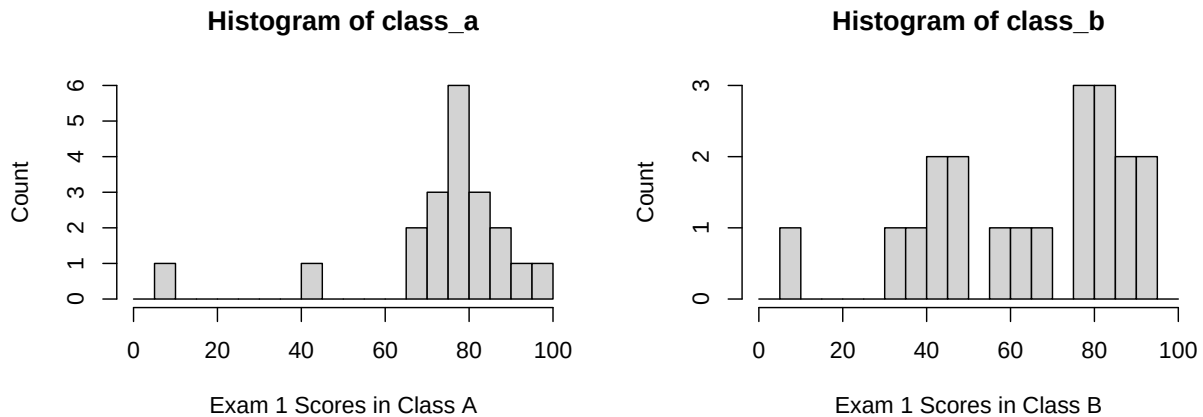


Figure 2.3: Scores on Exam 1 in Class A and Class B

The range of both distributions is (approximately) the same, but the spread looks very different. Only 2 out of 20 students scored below a 60 in Class A, while 8 students (out of 20) scored below a 60 in Class B.

2.2.2 Inter-Quartile Range (IQR)

Recall that the **median** divides the observations into two parts of equal size. This is because we defined the median as the value which half the observations are below and half are above. **Quartiles** divide the data into 4 equal parts. Specifically we define:

- The first quartile (or lower quartile), Q_1 , is the value such that 25% of the data lies below this value.
- The second quartile (or median), Q_2 , is value where 50% of the data lies below this value.
- The third quartile (or upper quartile), Q_3 , is the value such that 75% of the data lies below this value.

Computing the min, Q_1 , Q_2 , Q_3 , and max of a data set is known as the **five-number summary**.

Example 2.6. Below are the exact list of Exam 1 scores for Class A and Class B (from the histograms in Figure Figure 2.3). Compute the five-number summary for each class. What differences do you observe about the distribution from the statistics?

Class A	Class B
10	10
44	34
66	36
70	44
71	45
73	48
74	50
76	56
77	64
79	66
80	76
80	77
80	80
82	82
83	83
83	84
86	88
87	89

Class A	Class B
91	91
96	95

The **inter-quartile range** (IQR) is defined as $Q3 - Q1$.

Example 2.7. Compute the IQR for the exam scores by class.

! IQR Versus Range

Why might IQR be seen as a better gauge of spread than range?

2.2.3 Outliers

The general notion of an **outlier** is an observation that lies far away from the general distribution. The readings define outliers are points that are:

- Smaller than $Q1 - 1.5 \cdot (IQR)$, or
- Larger than $Q3 + 1.5 \cdot (IQR)$.

Do not think of this as a set in stone rule. In fact, the multiplier 1.5 was not chosen with any real mathematical theory behind it. Use these values to flag *potential* outliers in the dataset, but context is an important determining factor when understanding data.

Example 2.8. We have the following list of commute times for a neighborhood: 24, 25, 26, 27, 27, 28, 28, 29, 29, 30, 30, 30, 31, 31, 32, 32, 33, 33, 34, 34, 35, 35, 36, 36, 37, 38, 39, 40, 41, 42, 72, 74, 76, 78, 80, 82. Let's start by looking at the five-number summary.

```
[1] 24.0 29.5 33.5 39.5 82.0
```

From this we can compute the lower and upper fence:

$$\text{lower} = 29.5 - 1.5(10) = 14.5, \quad \text{upper} = 39.5 + 15 = 54.5$$

That means 6 data points are flagged as outliers using this rule: 72, 74, 76, 78, 80, 82. Let's look at the histogram of the distribution,

What we see from the histogram is a **bimodal** distribution. There is one subset of the neighborhood that commutes a shorter distance (think across town), while a small subset of the neighborhood commutes to a different city. Thus the 1.5 IQR outlier rule is a bad fit here.

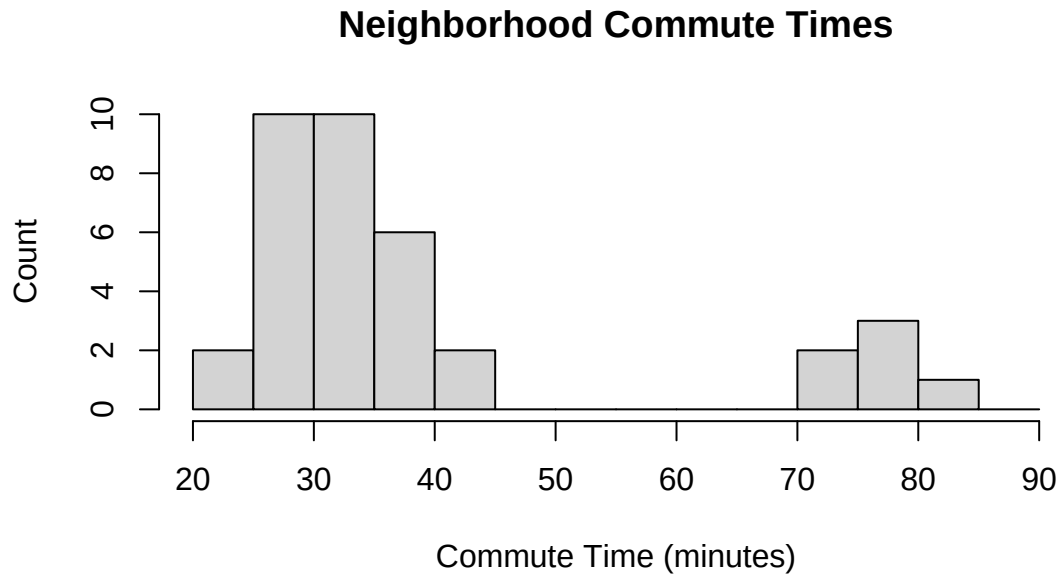


Figure 2.4: Histogram of neighborhood commute times.

! Removing Data

There is a false notion that outliers should be removed from the data set but this is **not true!** Outliers should be expected in data and are part of the underlying distribution. Only remove data points if they are the result of mistake in data entry or collection, or if they can be explained by a fundamentally different process which you are not interested in investigating.

2.3 Boxplots

Boxplots are a way to visualize the five number summary as well as outliers. Let's see this with an example.

Example 2.9 (Boxplots in StatCrunch). [For this example use this data on StatCrunch.](#) To make a boxplot of the data select Graph > Boxplot. Just select annual_income and then compute.

The box of the boxplot goes from Q1 to Q3 and the median is shown as a horizontal line. The two sticks (called whiskers) on either side of the box go until the largest data point *that isn't an outlier* and the stars (called fences) indicate outlier values. If we deselect the box indicating "Use fences to identify outliers" in StatCrunch then the whiskers will go all the way until the end of the range.

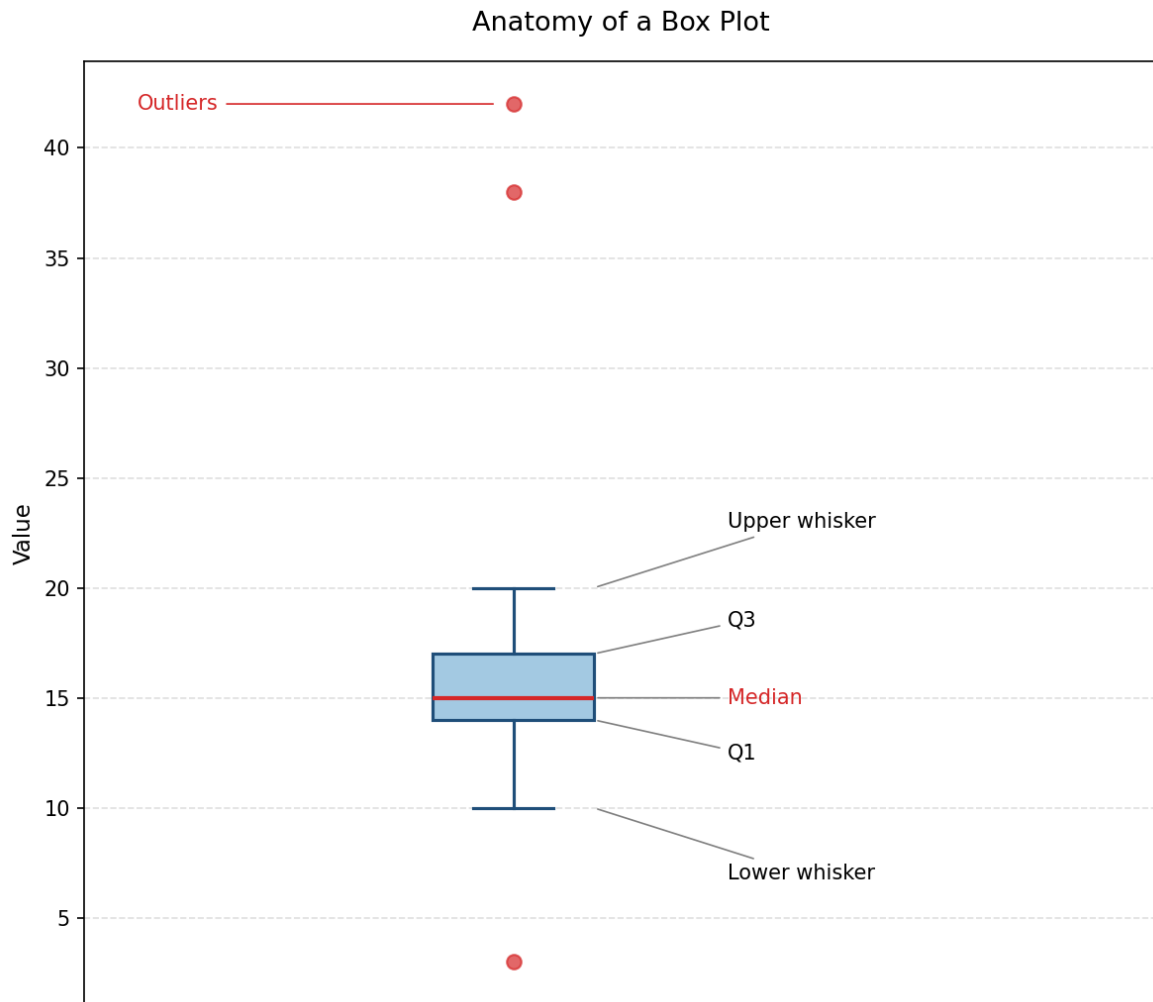


Figure 2.5: Labeled boxplot.

Example 2.10. The boxplots below show the typed words per minutes for support agents and software engineers. Compare the two distributions using what you know about boxplots.

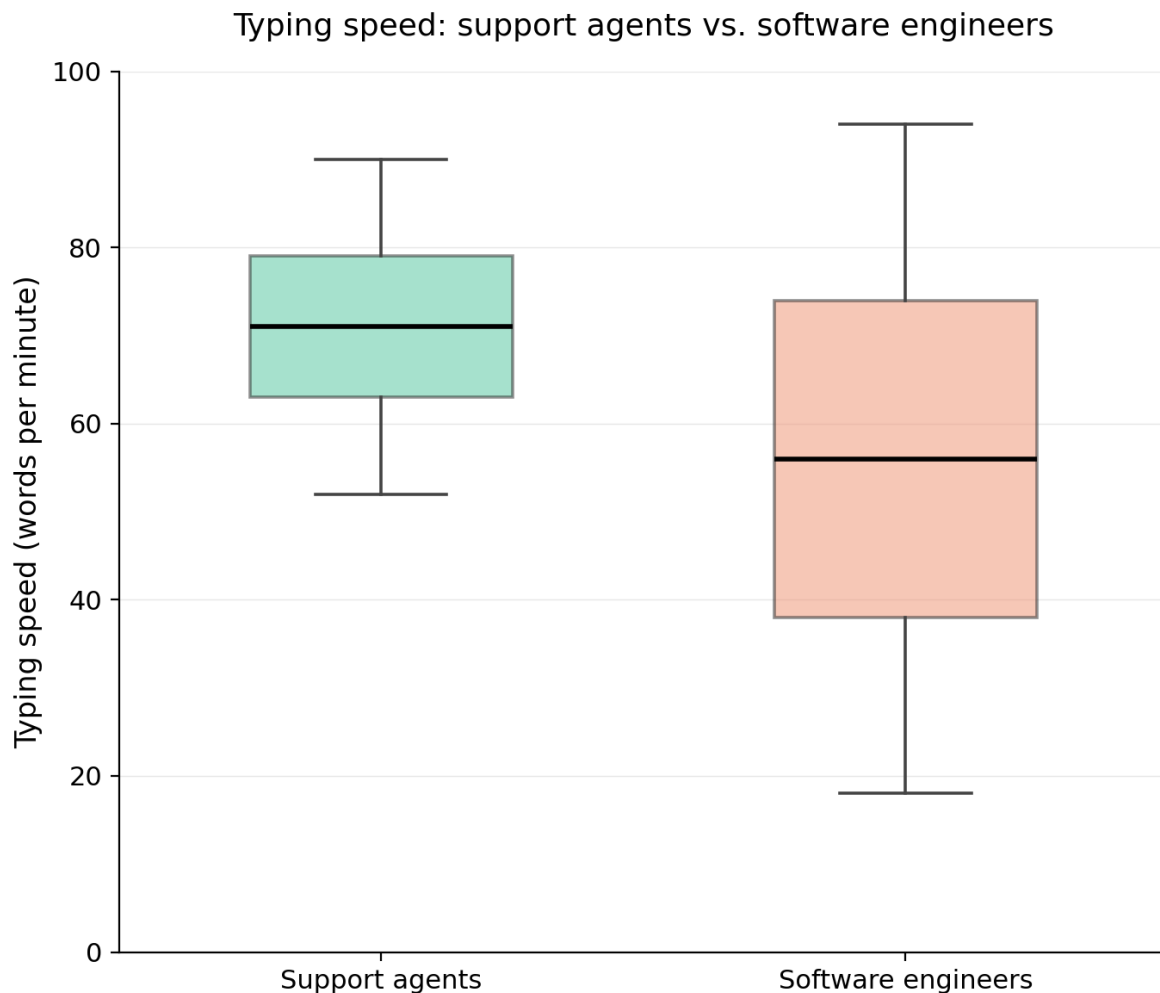


Figure 2.6: Typing speed boxplots for support agents and software engineers.

2.4 Standard Deviation

Here we will define and use the **sample standard deviation**. This is slightly different, but related to, the *population* standard deviation as well as the variance. These are terms we will define explicitly later. Given data points $x_1, \dots, x_n$ the **sample standard deviation** is defined by the formula

$$SD = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2}{n - 1}}$$

where $\bar{x}$ denotes the mean of the dataset. Let's see this formula in action with a small example,

Example 2.11. Anna denotes how many Reese's Peanutbutter Cups her 10 friends eat at a party. She gets the following amounts:

$$3, 4, 3, 6, 6, 2, 1, 8, 5, 5.$$

To compute the standard deviation, we first need to compute the mean,

$$\bar{x} = \frac{3 + 4 + 3 + 6 + 6 + 2 + 1 + 8 + 5 + 5}{10} = \frac{43}{10} = 4.3.$$

Next compute the differences,

$$-1.3, -0.3, -1.3, 1.7, 1.7, -2.3, -3.3, 3.7, 0.7, 0.7.$$

Square and sum,

$$1.69 + 0.09 + 1.69 + 2.89 + 2.89 + 5.29 + 10.89 + 13.69 + 0.49 + 0.49 = 40.1$$

Now divide by $n - 1 = 10 - 1 = 9$ and take the square root,

$$SD = \sqrt{\frac{40.1}{9}} = \sqrt{4.456} \approx 2.11.$$

We claim that standard deviation is telling us *something* about the spread of the distribution, but we can be more precise than that. The standard deviation tell us the average distance from an given datapoint to the mean. A larger standard deviation means the data is more spread out, while a smaller standard deviation tells us the data is compact.

Example 2.12 (Standard Deviation in StatCrunch). [For this example use this data on StatCrunch.](#) Compute the standard deviation of the income dataset.

Just like the mean, the sample standard deviation is heavily influenced by outliers.

2.5 Extra Practice

1. Compute the mean and median for the annual income dataset. Based on these values is the data skewed to the right or left? Generate a histogram of the data to confirm your thoughts. [Data](#)
2. A survey recorded the time spent on homework in two different grades. The five-number summaries are recorded below. Compute the IQR for each grade and sketch the boxplots (by hand). What are the main differences in the distributions?

	Grade 9	Grade 11
Minimum	60	40
Q1	90	70
Median	110	120
Q3	130	200
Maximum	160	260

3. The principal tells you that, based on the above summary data, 9th graders and 11th graders spend a similar amount of time on homework. How might you respond to the principal?
4. A group of 6 people were asked how many minutes they worked out for yesterday and the following results were obtained: 30, 60, 75, 0, 60, 45. Compute the mean and standard deviation by hand (calculator okay).
5. A news article reports: “The mean commute time is 25 minutes with a standard deviation of 40 minutes.” What does this suggest about the shape of the distribution? Why might this statement be misleading?

3 Examining Relationships

In this section we want to learn ways to examine the relationship between two variables. Typically, we are asking research questions about whether one variable impacts the other. The **response variable** or dependent variable is the variable that gives the outcome of the study. The **explanatory variable** is the variable that claims to impact the response variable.

Just like in Topic 1, these variables can be **quantitative** or **categorical**. In these notes we will spend some time looking at examples and the ways we can visualize and interpret relationships.

3.1 Side-by-Side Boxplots (C $\rightarrow$ Q)

If the explanatory variable is categorical and the response variable is quantitative, then one way to visualize the potential differences in distribution is by using side-by-side boxplots.

Example 3.1. Dataset. This dataset shows the race/ethnicity, age, and systolic BP for 360 patients (this data is simulated and not real). We would like to analyze if race/ethnicity impact patients BP.

To graph side-by-side boxplots in StatCrunch first select Graph > Boxplot. Next, select the column systolic_BP_mmHg and “group by” race_ethnicity.

Based on the graph, it looks like the race Black/African American has a higher Q1 and median than the other categories. We should be careful qualifying claims with just the above plots though!

! Important

What could be other reasons the population of black patients appear to have a higher systolic BP that isn't related to their race?

Another method of visualizing relationships between a categorical explanatory variable and a quantitative response variable is by using a stacked histogram. This can be done in StatCrunch as well.

Here we see that the histogram is colored by player position. From this visualization we can see that defenders typically don't score much.

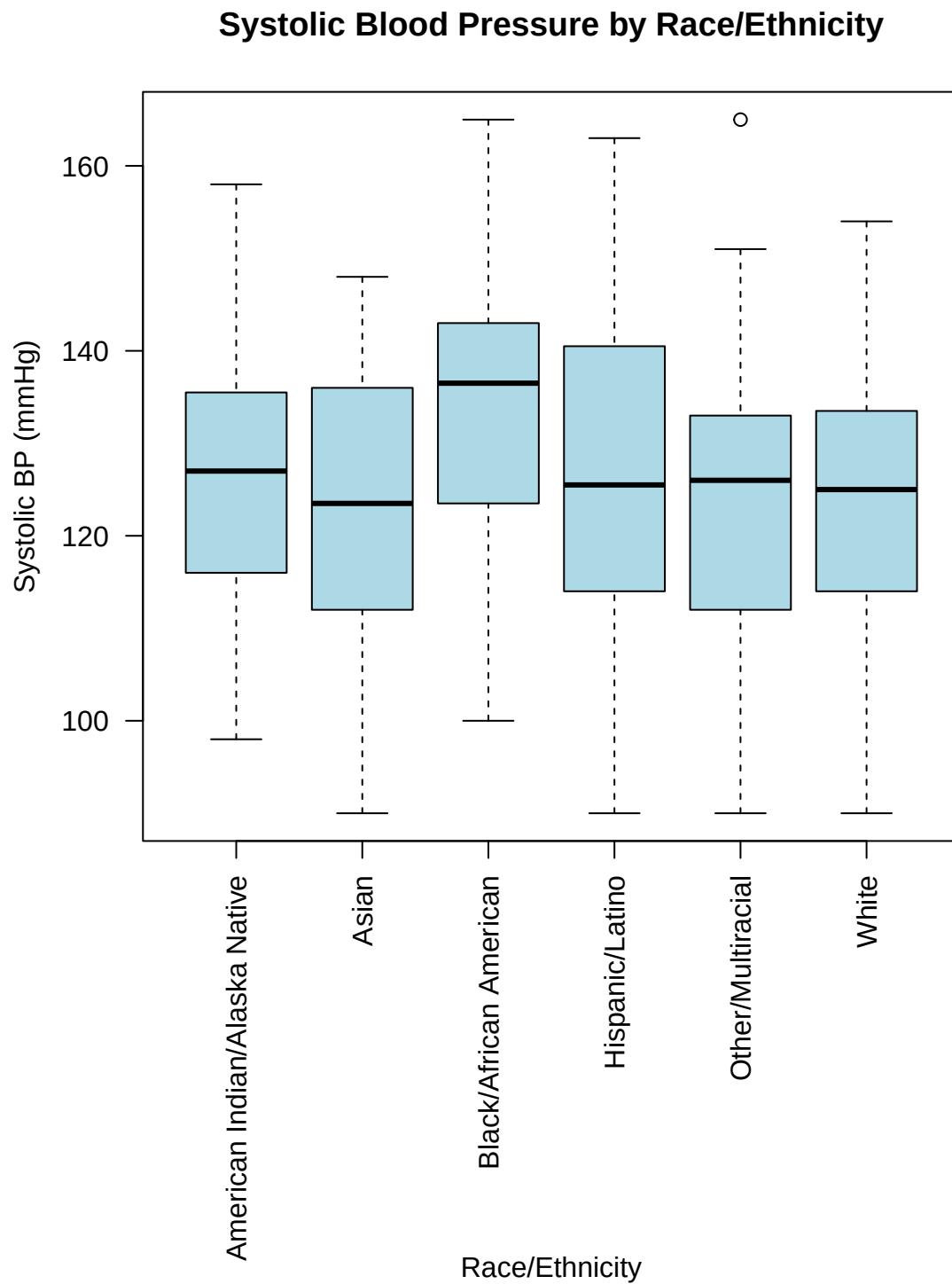


Figure 3.1: Side-by-side histogram of systolic BP.

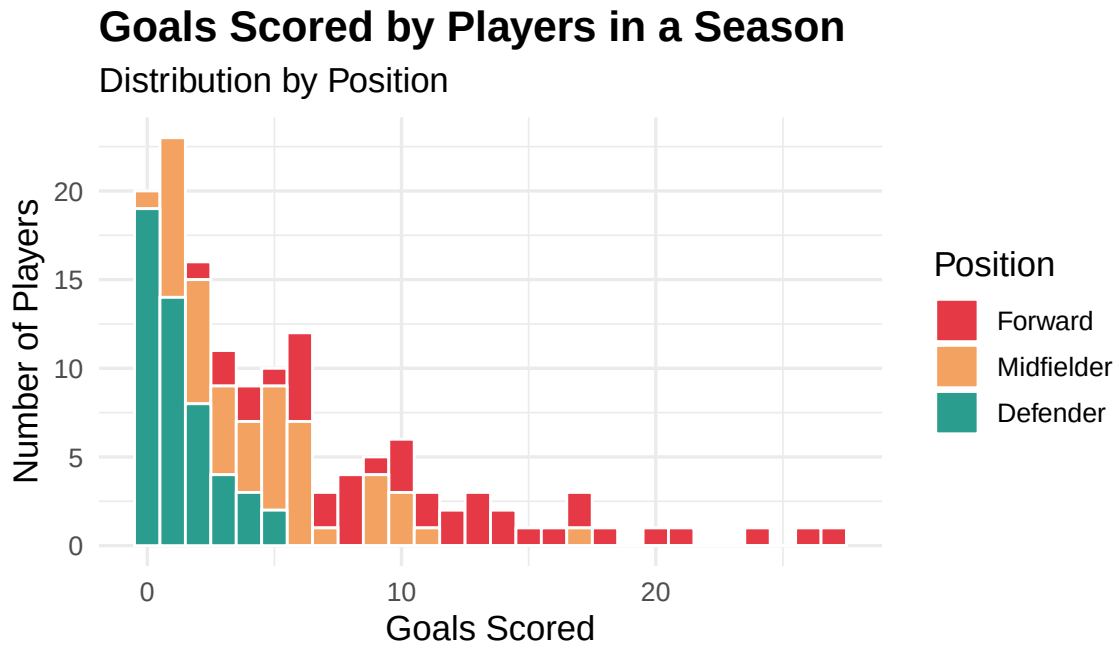


Figure 3.2: Stacked histogram of goals scored by different positions.

3.2 Contingency Tables (C $\rightarrow$ C)

3.3 Scatterplots (Q $\rightarrow$ Q)

Example 3.2. We want to determine if more time spent studying improves the score received on an exam. The amount of time spent studying was self reported by students. [Open the data in StatCrunch.](#)

Since we want to know if study time improves exam score, the explanatory variable is study time and the response variable is exam score. We can plot these using a scatter plot (Graph > Scatter Plot). We should set the study time as the x variable and the exam score as the y variable.

We see from the scatter plot that hours spent studying is **positively correlated** with exam scores. This means that the more hours spent studying, the higher of an exam score a student tends to receive. Notice that the points appear to show a “fuzzy” line with a positive slope.

Average hours of exercise weekly is **negatively correlated** with resting heart rate. This means the more hours exercised, the lower the resting heart rate.

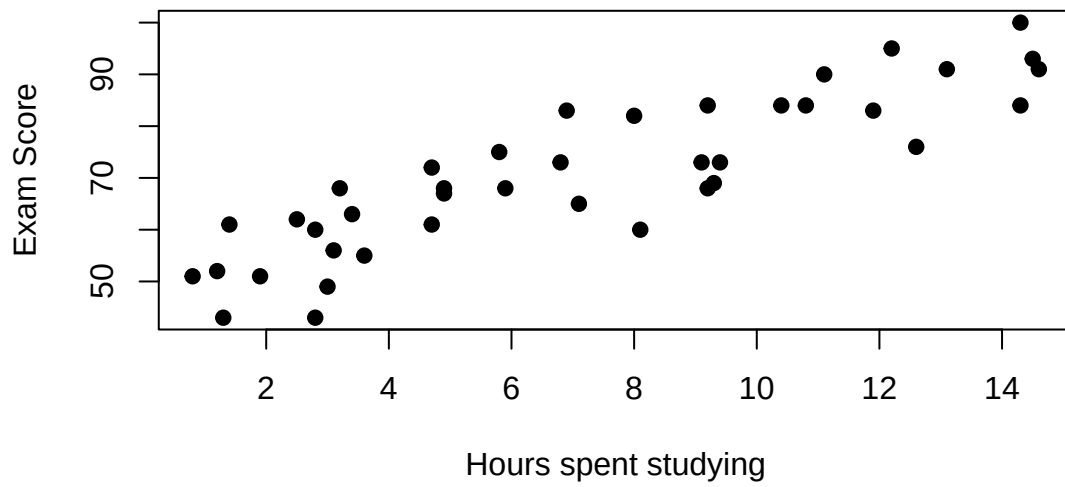


Figure 3.3: Scatter plot of time spent studying and exam scores.

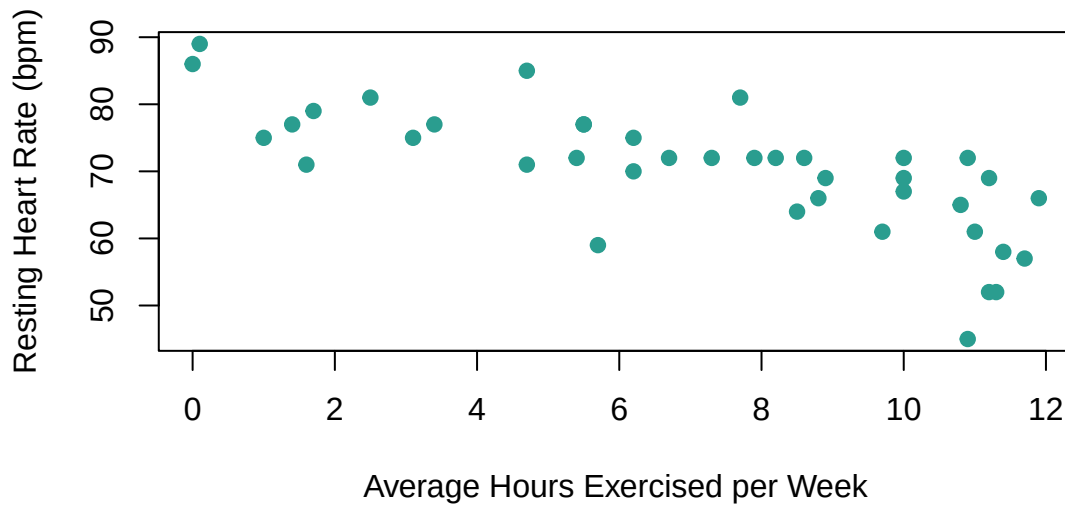


Figure 3.4: Scatter plot of weekly exercise hours and resting heart rate.